

Payroll Africa

Statutory payroll deduction automation for all 54 African countries,
built on ERPNext and Frappe HR.

GPL-3.0 · 54 countries · 111 reports · github.com/CoaleTech/payroll_africa

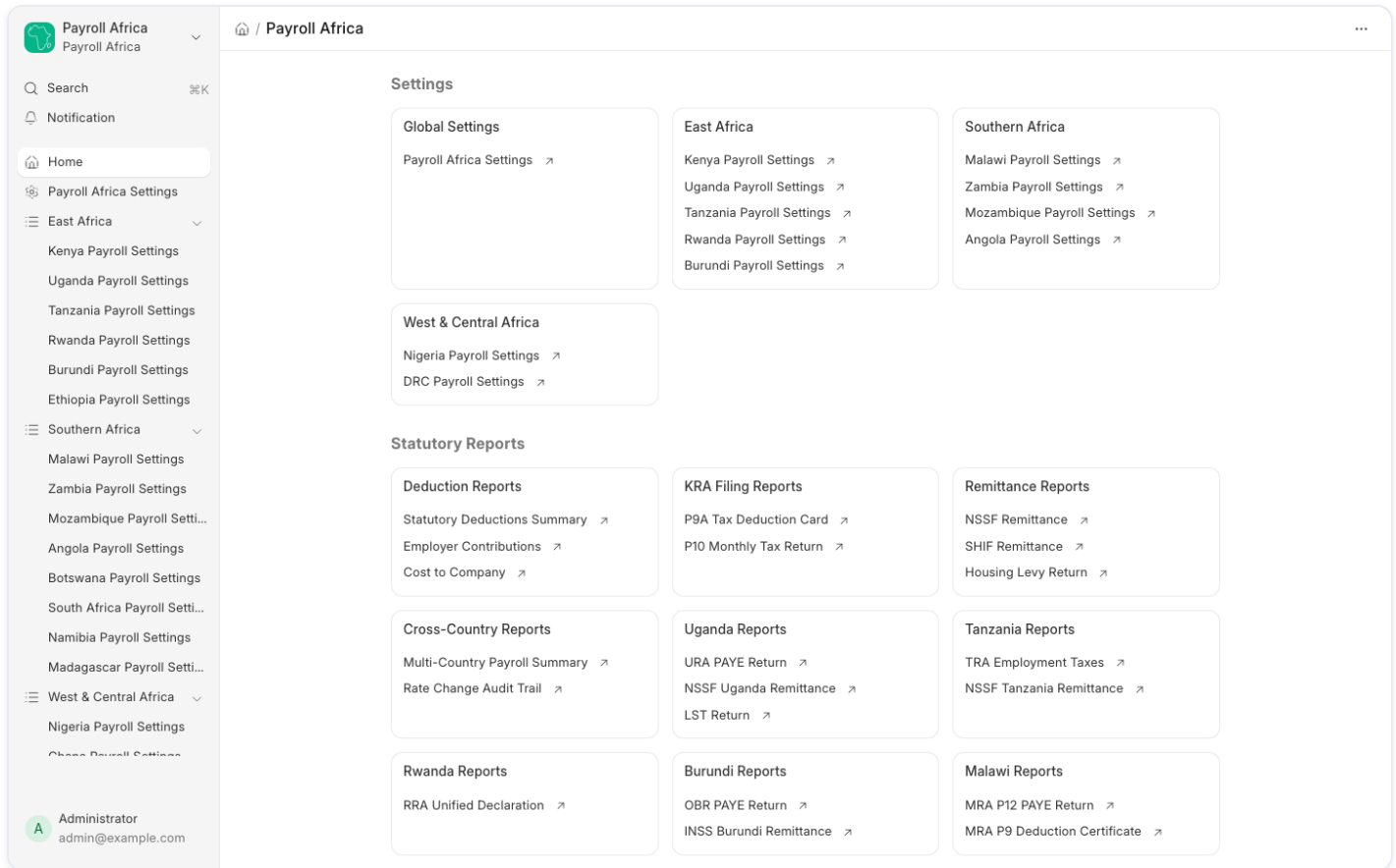
CONTENTS

1. [Introduction](#)
2. [Installation](#)
3. [Getting Started](#)
4. [Country Settings](#)
5. [Running Payroll](#)
6. [Reports](#)
7. [API Reference](#)
8. [Adding a New Country](#)

Introduction

Payroll Africa is an open-source app that automates statutory payroll deductions across **all 54 African countries** on ERPNext and Frappe HR. It computes employee and employer statutory contributions — PAYE, social security, health insurance, housing levies, training funds, and pension — on every Salary Slip, using each country's current tax bands and rates.

No manual calculation. No spreadsheets. Change a rate in settings, and the next payroll run picks it up.



Why Payroll Africa

Payroll teams across Africa maintain 54 different statutory rulebooks by hand, and those rates change every year. Getting them wrong means penalties from the revenue authority.

- Payroll Africa brings every country's statutory logic into one app, wired directly into the Frappe HR payroll engine.
- Rates and tax bands are configurable from the UI with effective dates — no code changes, no redeployment.
- Enable only the countries you operate in; the rest stay out of your way.

Key Features

- **Automatic deduction computation** on every Salary Slip via the Frappe HR `apply_regional_deductions` regional hook.
- **54 countries** — a dedicated calculator, configurable Settings page, and tax-band table for each.
- **111 reports** — country-specific PAYE returns and social-security remittances, plus cross-country summaries.

- **Country enable/disable** — toggle active countries from a single settings page; the sidebar and salary-component dropdowns update automatically.
- **Automatic country detection** — countries are enabled automatically from your Company records, on install and whenever a Company is added.
- **Rate-change tracking** — update statutory rates with effective dates, backed by an audit-trail report.
- **Salary structure templates & income tax slabs** — pre-built for every country with statutory deductions.
- **Standalone deduction API** — calculate deductions for any country without creating a Salary Slip.
- **Bulk recalculation** — recompute all enabled countries' draft slips in one call.

Supported Regions

Region	Countries
East Africa	Kenya, Uganda, Tanzania, Rwanda, Burundi, Ethiopia, Djibouti, Eritrea, Somalia, South Sudan, Sudan
Southern Africa	Malawi, Zambia, Mozambique, Angola, Botswana, South Africa, Namibia, Madagascar, Zimbabwe, Lesotho, Eswatini
West Africa	Nigeria, Ghana, Ivory Coast, Senegal, Mali, Niger, Burkina Faso, Benin, Togo, Guinea, Sierra Leone, Liberia, Gambia, Guinea-Bissau, Cabo Verde, Mauritania
North Africa	Egypt, Morocco, Tunisia, Algeria, Libya
Central Africa	DRC, Cameroon, Gabon, Congo, CAR, Chad, Equatorial Guinea, Sao Tome and Principe, Comoros
Indian Ocean	Mauritius, Seychelles, Madagascar

Under the Hood

- **Frappe Framework** — the full-stack Python/JavaScript web framework Payroll Africa is built on.
- **Frappe HR (HRMS)** — Payroll Africa extends the Salary Slip payroll engine through its supported regional-override hook, without patching core.
- **ERPNext** — provides the accounting, company, and currency foundations.

Next Steps

1. [Installation](#)
 2. [Getting Started](#)
 3. [Country Settings](#)
 4. [Running Payroll](#)
 5. [Reports](#)
 6. [API Reference](#)
 7. [Adding a New Country](#)
-

Installation

Prerequisites

- Frappe Framework v15 or v16
- ERPNext
- Frappe HR (HRMS)

Important: Payroll Africa computes deductions through Frappe HR's `apply_regional_deductions` regional hook on the Salary Slip. On HRMS versions that do not yet expose this hook, install a build that includes it (see the [upstream hook PR](#)). Without the hook, the calculators still work through the [standalone API](#), but deductions will not populate automatically on Salary Slips.

Install from the Frappe Cloud Marketplace

Add **Payroll Africa** to your bench from the Frappe Cloud Marketplace. ERPNext and Frappe HR are provisioned automatically as dependencies.

Install on a self-hosted bench

```
bench get-app https://github.com/CoaleTech/payroll_africa
bench --site your-site.local install-app payroll_africa
```

What installation provisions

On install, Payroll Africa sets up:

- **Currency records** for all required currencies.
- `Employee.payroll_country` custom Link field (to the standard Country DocType).
- **Country-specific Salary Components** for each deduction type.
- **Configurable Settings DocTypes** with current statutory tax bands for all 54 countries.
- **Income Tax Slabs and Salary Structure templates** for every country with statutory deductions (44 countries).
- **A workspace** with country sections and reports, plus a Desktop icon under Frappe HR.

Somalia and South Sudan have no mandatory statutory deductions (documented no-ops). Every other country exposes configurable rates and tax bands.

Verify the installation

1. Open the **Payroll Africa** workspace from the desk sidebar.
2. Confirm **Payroll Africa Settings** opens and lists the enabled countries.
3. Open a country settings page (e.g. **Kenya Payroll Settings**) and confirm its rate fields and tax bands are populated.

Uninstall

When the app is removed, its `before_uninstall` step cleans up the salary structures, salary components, income tax slabs, and custom fields it created.

```
bench --site your-site.local uninstall-app payroll_africa
```

Next Steps

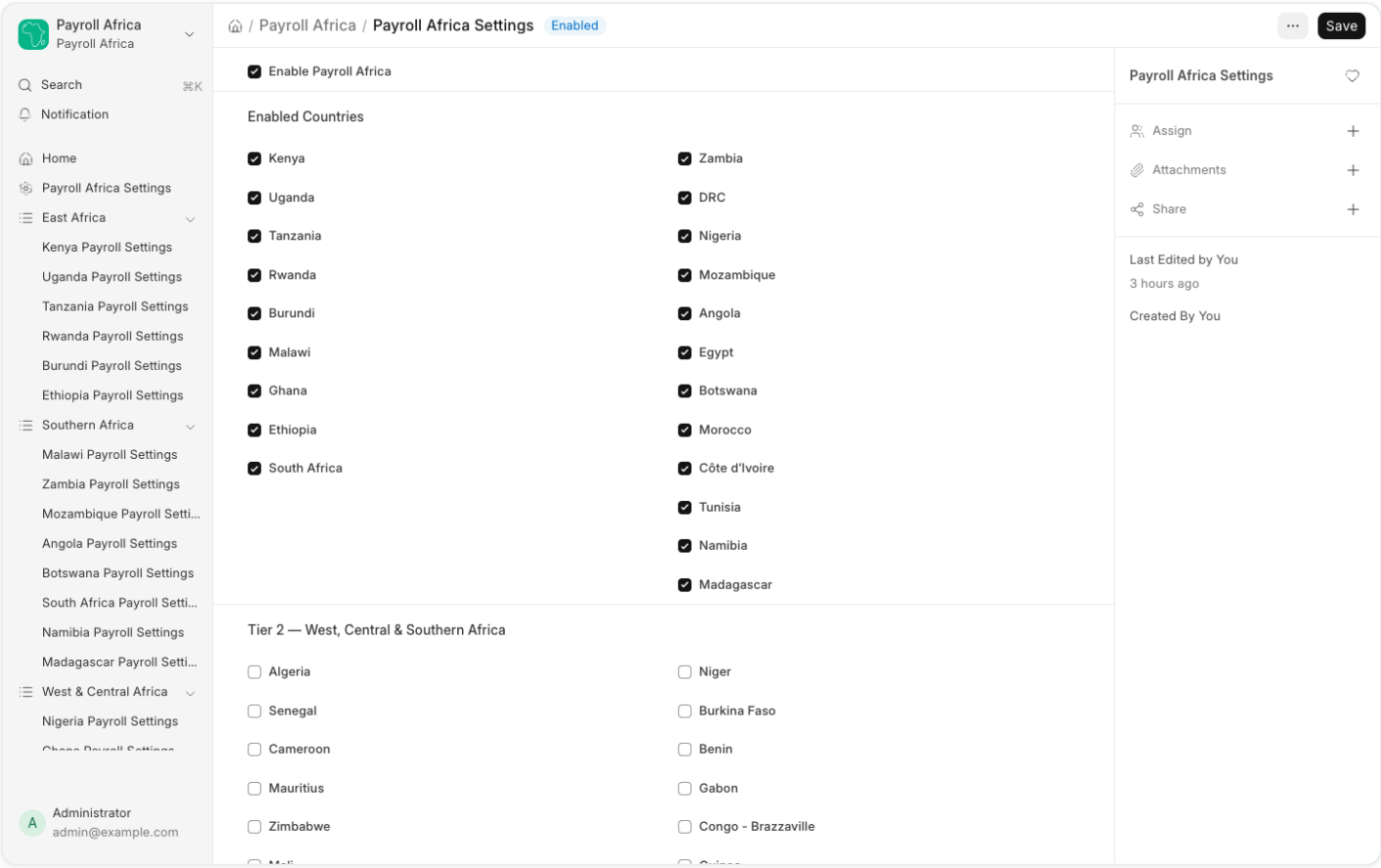
Continue to [Getting Started](#).

Getting Started

Payroll Africa is controlled from a single page: **Payroll Africa Settings**. Here you switch the app on and choose which countries are active.

Open Payroll Africa Settings

From the **Payroll Africa** workspace, click **Payroll Africa Settings** (under Global Settings), or search for it in the awesome bar.



Automatic country detection

Payroll Africa can enable countries for you automatically, so you often don't need to tick them by hand:

- **On installation**, it scans your existing Companies and enables the country of each one — a Company based in Kenya enables Kenya, a Company in Nigeria enables Nigeria, and so on.
- **When you add a new Company**, its country is enabled automatically.

This is **additive only**: it never disables a country you have turned off, and it does not change your selections when you run `bench migrate`. In the screenshot above, the ticked countries under **Enabled Countries** reflect the companies configured on this site, plus any manual choices. You can always override the result in the next section.

Enable the app

Tick **Enable Payroll Africa** at the top of the page. This is the master switch — when it is off, no statutory deductions are computed for any country.

Enable countries

Under **Enabled Countries**, tick each country you operate payroll in. Countries are grouped by region and tier so you can find them quickly.

When a country is **enabled**:

- Its settings page and reports appear in the workspace sidebar.
- The engine computes its statutory deductions on matching employees' Salary Slips.
- Its salary components are available in Salary Structure dropdowns.

When a country is **disabled**:

- Its settings and report links disappear from the sidebar immediately — no restart required.
- The engine silently skips employees in that country.
- Its salary components are hidden from Salary Structure component dropdowns.

Save

Click **Save**. Changes take effect on the next Salary Slip computation — there is nothing to migrate or restart.

Next Steps

- Configure each active country's rates in [Country Settings](#).
- Set up employees and run payroll in [Running Payroll](#).

Country Settings

Every enabled country has its own Settings page in the workspace sidebar (for example **Kenya Payroll Settings**). This is where you keep statutory rates and tax bands current — no code changes required.

Payroll Africa

Payroll Africa

Search

Notification

Home

Payroll Africa Settings

East Africa

- Kenya Payroll Settings
- Uganda Payroll Settings
- Tanzania Payroll Settings
- Rwanda Payroll Settings
- Burundi Payroll Settings
- Ethiopia Payroll Settings

Southern Africa

- Malawi Payroll Settings
- Zambia Payroll Settings
- Mozambique Payroll Setti...
- Angola Payroll Settings
- Botswana Payroll Settings
- South Africa Payroll Setti...
- Namibia Payroll Settings
- Madagascar Payroll Setti...

West & Central Africa

- Nigeria Payroll Settings
- Chad Payroll Settings

Administrator

admin@example.com

Payroll Africa / Kenya Payroll Settings

Enabled

...

Save

Effective From

2025-01-01

Enabled

PAYE

Monthly Personal Relief

2,400.00

Monthly personal tax relief (KES)

Insurance Relief Rate

15.00

Percentage of insurance premiums eligible for relief

Insurance Relief Cap

5,000.00

Maximum monthly insurance relief (KES)

PAYE Bands

PAYE Bands

Progressive PAYE tax bands (monthly amounts in KES)

No.	From Amount *	To Amount *	Rate (%) *	
1	ZK 0.00	ZK 24,000.00	10.00%	
2	ZK 24,001.00	ZK 32,333.00	25.00%	
3	ZK 32,334.00	ZK 500,000.00	30.00%	
4	ZK 500,001.00	ZK 800,000.00	32.50%	
5	ZK 800,001.00	ZK 0.00	35.00%	

Add row

NSSF

Tier I Rate

6.00

Tier II Rate

6.00

Kenya Payroll Settings

Assign

Attachments

Share

Last Edited by You

yesterday

Created By You

Common fields

Each country settings page includes:

- **Effective From** — the date the current rates took effect. Update it whenever you change rates, so the change is tracked.
- **Enabled** — disable a single country here without touching the global switch in Payroll Africa Settings.
- **Rate fields** — contribution percentages, ceilings, minimum wages, and reliefs specific to the country.
- **Tax band table** — the progressive tax brackets (from-amount, to-amount, rate). Edit brackets directly in the grid.

Tax bands

Progressive taxes (PAYE, PIT, IRPP, etc.) are defined as bands in a child table. Each row has a lower bound, an upper bound, and a rate. Set the top band's upper bound to 0 to mean "and above".

For example, Kenya's PAYE bands and NSSF tiers, SHIF rate, Housing Levy, NITA, and personal relief are all editable on the Kenya Payroll Settings page.

Examples of country-specific settings

Country	Configurable fields
Kenya	NSSF Tier I/II rates and caps, SHIF rate, Housing Levy rates, NITA, personal relief, PAYE bands
South Africa	PAYE rebates, UIF rate and annual cap, SDL rate
Zimbabwe	PAYE bands (USD or ZiG), NSSA rate and annual ceiling, AIDS Levy rate, currency mode
Senegal	IPRES rates and ceiling, CSS Health rates, AMO Health toggle, income-tax bands, family deductions
Mauritius	NSF rate and ceiling, CSG rate, HRDC rate, PRGF toggle, Fair Share Contribution toggle
Cameroon	CNPS rates, family allowances, work-injury risk class, CFC housing, FNE rate, PIT abatement, CRTV bands, Taxe Communale bands

Fallback defaults

If a country's Settings page has not been customized, its calculator falls back to current statutory defaults, so payroll works correctly immediately after installation. Configure the page to override those defaults with your organization's exact figures.

Next Steps

Continue to [Running Payroll](#).

Running Payroll

Once a country is enabled and configured, Payroll Africa computes its statutory deductions automatically whenever you create a Salary Slip for an employee in that country.

How country is resolved

For each Salary Slip, the employee's payroll country is resolved in this order:

1. **Salary Structure Assignment** → Payroll Country
2. **Employee** → Payroll Country
3. **Company** → Country

Set the **Payroll Country** field on the Employee (or on the Salary Structure Assignment for date-effective changes). If it is not set, the engine falls back to the company's country.

Set up an employee

1. Open the **Employee** record.
2. Set **Payroll Country** to the relevant country (e.g. Kenya).
3. Assign a **Salary Structure** — use the country's pre-built template (e.g. "Kenya Payroll Template") or your own.

You do not need to add statutory components manually; the calculator appends any missing ones during Salary Slip validation.

Create a Salary Slip

Create and save a Salary Slip for the employee as usual. Payroll Africa runs during computation and populates:

- **Employee deductions** in the Salary Slip Deductions table.
- **Employer-only contributions** in the Employer Contributions table (when present).

The example below shows the same employee before and after Payroll Africa is active.

Before — with Payroll Africa disabled, the Salary Slip shows only structure components; the Deductions table is empty.

Payroll

Frappe HR

Search

Notification

Home

Dashboard

Payroll Entry

Salary Structure Assignm...

Salary Slip

Additional Salary

Salary Withholding

Reports

Setup

Settings

Administrator

admin@example.com

Payroll / Salary Slip / PA Demo Employee

Draft

Submit this document to confirm

Details

Payment Days

Earnings & Deductions

Net Pay Info

Bank Details

PA Demo Employee

Sal Slip/HR-EMP-00476...

Assign

Attachments

Tags

Share

Last Edited by You

6 minutes ago

Created By You

6 minutes ago

Earnings

Component *	Amount (IN...
PA Demo Basic	₹ 200,000.00

Add row

Deductions

Component *	Amount (IN...
No rows	

Add row

Totals

Gross Pay (INR)

₹ 200,000.00

Gross Year To Date (INR)

₹ 200,000.00

Total Deduction (INR)

₹ 0.00

Comments

A

Type a reply / comment

Activity

You created this · 6 minutes ago

You last edited this · 6 minutes ago

+ New Email

After — with Payroll Africa enabled, the Kenya statutory rows — NSSF Employee, SHIF, Housing Levy, NSSF Employer, Employer Housing Levy, and NITA — are appended automatically, and the totals update.

Payroll

Frappe HR

Search

Notification

Home

Dashboard

Payroll Entry

Salary Structure Assignm...

Salary Slip

Additional Salary

Salary Withholding

Reports

Setup

Settings

Administrator

admin@example.com

Payroll / Salary Slip / PA Demo Employee

Draft

Submit this document to confirm

Details

Payment Days

Earnings & Deductions

Net Pay Info

Bank Details

PA Demo Employee

Sal Slip/HR-EMP-00476...

Assign

Attachments

Tags

Share

Last Edited by You

6 minutes ago

Created By You

6 minutes ago

Earnings

Component *	Amount (IN...
PA Demo Basic	₹ 200,000.00

Add row

Deductions

Component *	Amount (IN...
NSSF Employee	₹ 6,480.00
SHIF	₹ 5,500.00
Housing Levy	₹ 3,000.00
NSSF Employer	₹ 6,480.00
Employer Housing Levy	₹ 3,000.00
NITA	₹ 50.00

Add row

Totals

Gross Pay (INR)

₹ 200,000.00

Gross Year To Date (INR)

₹ 200,000.00

Total Deduction (INR)

₹ 14,980.00

Comments

Bulk recalculation

After changing rates, recompute existing **draft** Salary Slips without reopening each one:

- **One country:** `recalculate_salary_slips(country, from_date, to_date, company)`
- **All enabled countries:** `recalculate_all_countries(from_date, to_date)`

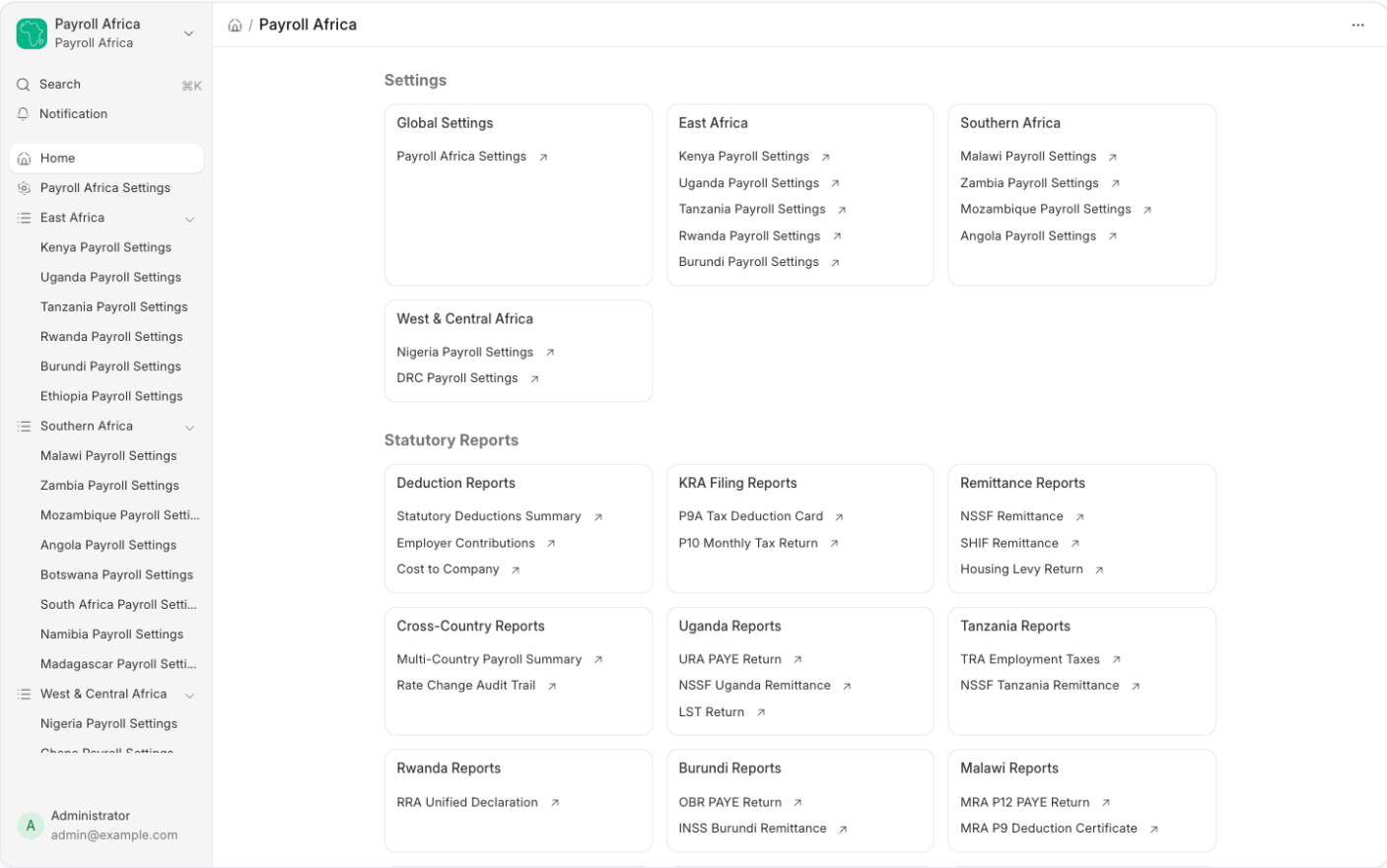
See the [API Reference](#) for details. Submitted slips are never modified.

Next Steps

Continue to [Reports](#).

Reports

Payroll Africa ships **111 reports** — country-specific statutory returns and remittances, plus cross-country summaries. All reports are available from the **Statutory Reports** section of the Payroll Africa workspace.



Cross-country reports

Report	Description
Statutory Deductions Summary	All employee deductions by period, with dynamic columns per component
Employer Contributions	Employer-only statutory costs per employee
Cost to Company	Total compensation including all employer contributions
Multi-Country Payroll Summary	Consolidated view across all enabled countries
Rate Change Audit Trail	History of statutory rate changes with effective dates

Country reports

Each enabled country provides its own statutory returns and remittance reports. A few examples:

Country	Reports
Kenya	P9A Tax Deduction Card, P10 Monthly Tax Return, NSSF Remittance, SHIF Remittance, Housing Levy Return
Uganda	URA PAYE Return, NSSF Uganda Remittance, LST Return
Nigeria	PAYE Schedule, PenCom Remittance, NHF Remittance, NHIS Schedule
Ghana	PAYE Schedule, SSNIT Remittance
South Africa	PAYE Remittance, UIF Remittance
Egypt	Income Tax Return, Social Insurance Remittance

Running a report

1. Open the report from the **Statutory Reports** section of the workspace, or search for it.
2. Set the filters — typically **Company** and a **date range** (and country, for cross-country reports).
3. The report renders once Salary Slips exist in the selected period. Export to Excel or CSV from the report menu.

Next Steps

Continue to the [API Reference](#).

API Reference

All endpoints are `@frappe.whitelist()` decorated and accessible over REST.

`calculate_deductions`

Compute deductions for any country without creating a Salary Slip — useful for simulations and what-if analysis.

```
from payroll_africa.api import calculate_deductions

result = calculate_deductions("Kenya", gross_pay=100000)
# {
#   "country": "Kenya",
#   "gross_pay": 100000.0,
#   "deductions": [
#     {"component": "NSSF Employee", "amount": 2160.0, "is_employer_only": False},
#     {"component": "NSSF Employer", "amount": 2160.0, "is_employer_only": True},
#     ...
#   ],
#   "employee_total": 6035.0,
#   "employer_total": 3210.0,
#   "net_pay": 93965.0,
#   "cost_to_company": 103210.0
# }
```

`get_supported_countries`

Return the sorted list of all 54 supported country names.

`recalculate_salary_slips`

Recalculate all **draft** Salary Slips for a single country in a date range.

```
from payroll_africa.api import recalculate_salary_slips
```

```
result = recalculate_salary_slips("Kenya", "2025-01-01", "2025-12-31", company="My Co")
# {"updated": 42, "errors": [], "message": "42 salary slip(s) recalculated"}
```

recalculate_all_countries

Recalculate draft slips for every enabled country in one call.

```
from payroll_africa.api import recalculate_all_countries

result = recalculate_all_countries("2025-01-01", "2025-12-31")
# {
#   "updated": 142,
#   "by_country": {"Kenya": {...}, "Uganda": {...}, ...},
#   "message": "142 salary slip(s) recalculated across 5 countries"
# }
```

REST examples

```
curl -X POST https://your-site/api/method/payroll_africa.api.calculate_deductions \
-H "Authorization: token <api_key>:<api_secret>" \
-d '{"country": "Kenya", "gross_pay": 100000}'

curl -X POST https://your-site/api/method/payroll_africa.api.recalculate_all_countries \
-H "Authorization: token <api_key>:<api_secret>" \
-d '{"from_date": "2025-01-01", "to_date": "2025-12-31"}'
```

Next Steps

Developers extending Payroll Africa should read [Adding a New Country](#).

Adding a New Country

Payroll Africa is designed to be extended. Adding a country follows a consistent pattern.

Steps

1. **Create the calculator** — `calculators/{snake}.py` extending `BaseCalculator`, implementing `compute()` and returning `{component: {amount, is_employer_only}}`.

2. **Create the Settings DocType** — `{country}_payroll_settings` (Single) with rate fields and a tax-band child table.
3. **Create the Band DocType** — `{country}_paye_band` or `{country}_pit_band` (child table: `from_amount`, `to_amount`, `rate`).
4. **Register** — add entries to `COUNTRY_MAP` and `SETTINGS_MAP` in `engine/registry.py`.
5. **Boot** — add `"Country Name": "enable_{snake}"` to `COUNTRY_FIELD_MAP` in `boot.py`.
6. **Settings toggle** — add an `enable_{snake}` Check field to `payroll_africa_settings.json`.
7. **Sidebar** — add the settings link to `workspace_sidebar/payroll_africa.json`.
8. **Reports** — create the tax-return and social-security remittance reports under `payroll_africa/report/`.
9. **Tests** — add a test class covering zero gross, a rate spot-check, employer flags, ceilings, and PIT thresholds.
10. **Migrate** — run a site migration to install the new DocTypes.

The calculator pattern

Each country implements `BaseCalculator` :

```
class KenyaCalculator(BaseCalculator):
    def compute(self, salary_slip) -> dict:
        gross = self.get_gross_pay(salary_slip)
        results = {}

        nssf = self._compute_nssf(gross)
        results["NSSF Employee"] = {"amount": nssf["employee"], "is_employer_only": False}
        results["NSSF Employer"] = {"amount": nssf["employer"], "is_employer_only": True}
        # ... SHIF, Housing Levy, NITA ...

        return results
```

The registry maps country names to calculator classes and loads their settings:

```
# engine/registry.py
COUNTRY_MAP = {
    "Kenya": "payroll_africa.calculators.kenya.KenyaCalculator",
    "Senegal": "payroll_africa.calculators.senegal.SenegalCalculator",
    # ... 54 entries
}
```

Calculators fall back to hardcoded statutory defaults when the Settings DocType is empty, so payroll runs correctly out of the box.

Testing

Run the suite before submitting:

```
bench --site your-site.local run-tests --app payroll_africa
```

Every country (except the documented Somalia/South Sudan no-ops) is covered by unit tests.